

Over time, these components will combine into increasingly sophisticated trading systems that can analyze markets, identify opportunities, and execute strategies with precision and discipline.

In the next chapter, we'll explore data structures—the various ways Python can organize information like price histories, portfolio positions, and market data. These structures will become the backbone of your trading systems, working hand-in-hand with the functions you've learned to create powerful algorithmic strategies.

Remember that every great trading system, from the simple to the sophisticated, is ultimately built from well-designed functions. Master this foundation, and you'll be well on your way to developing effective algorithmic trading strategies of your own.

*?? **Pro Tip:** Professional algorithmic traders distinguish between "research functions" and "production functions." Research functions prioritize flexibility and readability for exploring ideas, while production functions prioritize reliability and performance for actual trading. As you develop your own trading systems, maintain separate function libraries for these different purposes. This dual approach, used by firms like AQR Capital and Two Sigma, accelerates strategy development while ensuring that only thoroughly tested components make it into live trading environments.*

Project 7: Technical Analysis Function Library

Objective: Build a comprehensive technical analysis function library that can serve as the foundation for various trading strategies. This project will help you implement, organize, and test the core building blocks of a professional trading system.

Requirements:

1. Create modular functions for at least 6 different technical indicators, including: